

Pushkar Singhania

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Summary

Electrical and Computer Science undergraduate at VIT Chennai (CGPA: 9.20/10) with strong foundations in **Java, Python, Data Structures, Algorithms, Object-Oriented Programming, SQL, and Software Engineering**. Skilled in building **full-stack web applications, REST APIs, backend systems, and machine learning solutions**. Experienced with collaborative development, problem solving, debugging, version control, and scalable application design.

Education

Vellore Institute of Technology, Chennai	2024 – 2028
B.Tech in Electrical and Computer Science Engineering	CGPA: 9.20 / 10
GD Birla Centre for Education, Kolkata	Class XII (2024)
Higher Secondary Education	92%
Sacred Heart School	Class X (2022)
Secondary Education	88%

Technical Skills

- **Languages:** Java, Python, C, JavaScript
- **Backend & APIs:** REST APIs, Node.js, Express, FastAPI
- **Databases:** SQL, MySQL
- **Frontend:** HTML, CSS, JavaScript, React, Tailwind CSS
- **Machine Learning:** TensorFlow, NumPy, Pandas
- **Electrical & Embedded:** Embedded Systems, Microcontrollers (8051, Arduino)
- **Tools & Platforms:** Git, GitHub, Linux, VS Code, MATLAB, Keil uVision

Projects

FinSight – Financial Document Analysis Platform *React, Node.js, Express*

- Engineered a full-stack web platform for financial document processing and structured data visualization.
- Integrated OCR-based extraction workflows and implemented REST APIs using Node.js and Express.
- Created reusable frontend components and responsive layouts using React and Tailwind CSS.

DoseWise – Insulin Management System *React, FastAPI, PostgreSQL*

- Built a healthcare management platform for insulin tracking and glucose data analysis.
- Implemented backend APIs using FastAPI for handling user workflows and health-related data.
- Structured relational database schemas in PostgreSQL for secure and organized data storage.

AI-Based Fault Detection in Electrical Machines *Python, Scikit-learn*

- Created a machine learning pipeline to classify electrical machine faults using sensor datasets.
- Applied preprocessing, feature extraction, and classification techniques using Python and Scikit-learn.

Drowsiness Detection Smart Glasses *Arduino, Embedded C*

- Constructed a wearable embedded system for driver drowsiness monitoring using sensors and Arduino.
- Configured alert mechanisms through microcontroller programming and real-time signal analysis.
- Tested hardware response and system reliability under multiple operating conditions.

Audio Signal Processing and Analysis *MATLAB, DSP*

- Analyzed audio signals using FFT, filtering, and waveform visualization techniques in MATLAB.
- Performed frequency-domain analysis and signal enhancement for multiple audio inputs.

Experience

Media and Creative Head *IE(I) Student Chapter, VIT Chennai*

- Led media initiatives for **5+ technical workshops and events**, coordinating cross-functional teams to deliver digital content.
- Managed promotional campaigns and social media outreach to improve student engagement and event visibility.

Board Member *Mharo Rajasthan Club, VIT Chennai*

- Organized multiple student events and cultural programs for audiences of **100+ students**.
- Coordinated logistics and collaborated with teams to improve event execution and participation.

Leadership & Achievements

- **School Prefect** — GD Birla School; selected among top students for leadership, responsibility, and communication skills.